

① $y' = x$

$$y = \frac{x^2}{2} + C$$

② $y' = y \quad / : y$

$$\frac{y'}{y} = 1 \rightarrow \log(y) = x + C_1 \rightarrow \boxed{y = \exp(x) \cdot C_2}$$

"ez egy katasztrófa"

Van-e más megoldás? A megoldások konstansban különb. csak (?)

$$(y e^{-x})' = \underbrace{y'}_y e^{-x} + y(-e^{-x}) = 0 \rightarrow y e^{-x} = C \rightarrow y = C e^x$$

③ $x y' + 2y = 0 \rightarrow x y' = -2y \rightarrow y' = -\frac{2y}{x} \rightarrow \frac{y'}{y} = -\frac{2}{x} \rightarrow$

$$\rightarrow \log(y) = -2 \log(x) + C_1 \rightarrow y = (e^{\log x})^{-2} C_2 = x^{-2} C_2$$

④ $x y' + 2y = 3x^2$ LDE $(a(x)y' + b(x)y = c(x))$
homogén, ha $c(x) = 0$

Ma előző homogén rész

$$x y' + 2y = 0 \rightarrow y_h = x^{-2} C_2$$

az eredeti inhomogén LDE-t 3féleképp lehet megoldani

i) Ha már y' el-ja 1, akkor osztunk a hom. m.-sal*

$$x y' + 2y = 3x^3 \quad / : x$$

$$y' + \frac{2}{x} y = 3x^2 \quad / : \frac{1}{x^2}$$

$$x^2 y' + 2x y = \frac{3}{5} x^5 + C \quad / \int \dots$$

$$x^3 y = \frac{3}{5} x^5 + C$$

$$y = \frac{3}{5} x + \frac{C}{x^2}$$

partikuláris

$$y = \underbrace{y_p}_{\text{partikuláris}} + y_h$$

* C_2 nélkül

ii) „Állandó variálása”

Keressük a most $y(x) = c(x) \cdot y_h(x)$ alakban

$$\text{Most } y = c(x) \cdot \frac{1}{x^2} \rightarrow y' = c'(x)x^{-2} + c(x)(-2)x^{-3}$$

Behelyettesítve:

$$x y' + 2y = 3x^3$$

$$c'(x) \cdot x^{-1} + \underbrace{c(x) \cdot (-2)x^{-2} + 2c(x) \cdot x^{-2}}_0 = 3x^3$$

$$c'(x) = 3x^4$$

$$c(x) = \frac{3}{5}x^5 + c_2$$

$$x^2 y = \frac{3}{5}x^5 + c_2 \rightarrow y = \frac{3}{5}x^3 + \frac{c_2}{x^2}$$

iii) próbafüggvény

$$y_p = A x^3$$

próba/particularis

$$y_p' = 3A x^2$$

$$x y_p' + 2y_p = 3A x^3 + 2A x^3 = 5A x^3 = 3x^3 \rightarrow$$

$$\rightarrow A = \frac{3}{5} \rightarrow y_p = \frac{3}{5}x^3$$

$$\text{és } y = y_h + y_p = \frac{c}{x^2} + \frac{3}{5}x^3$$

⑤ $y' - \frac{1}{x \log(x)} y = x \log(x)$

$$y' - \frac{y}{x \log(x)} = 0 \rightarrow y' = \frac{y}{x \log(x)} \rightarrow \frac{y'}{y} = \frac{1}{x \log(x)} \int \dots$$

$$\log(y) = \log(\log(x)) + c$$

$$y_h = \log(x) \cdot c_2$$

$$3. \frac{y}{\log(x)} - \frac{y}{x \log^2(x)} = x$$

$$\left(y \cdot \frac{1}{\log(x)} \right)' = x$$

$\uparrow$
 y_h

$$y \cdot \frac{1}{\log(x)} = \frac{x^2}{2} + C_2 \rightarrow y = \frac{x^2 \log(x)}{2} + C_2 \log(x)$$

$$⑥. y' - \frac{y}{x} - 2x^2 = 0 \rightarrow y' - \frac{y}{x} = 2x^2 \quad y(1) = -1$$

$$y' - \frac{y}{x} = 0 \rightarrow \frac{y'}{y} = \frac{1}{x} \rightarrow \log(y) = \log(x) + C \rightarrow y_h = Cx$$

$$\underbrace{\frac{y'}{x} - \frac{y}{x^2}}_{\frac{xy' - y}{x^2}} = 2x \rightarrow \frac{y}{x} = \frac{2x^2}{2} + C_2$$

$$y = x^3 + C_2 x$$

$$y(1) = 1 + C_2 \rightarrow C_2 = -2$$

$$y = x^3 - 2x$$

$$\left(\frac{y}{x} \right)'$$

$$⑦. y' - 2y = x - 2 \quad y(0) = 3$$

H# megoldani

Szükségszerű approximáció

1. integráljuk a kép-ből

Kieserjük x -et valami másra, majd int.-unk 0-tól x -ig

$$\int_0^x y'(s) ds = \int_0^x (2y(s) + s - 2) ds$$

$\xrightarrow{\text{kezdeti feltételből}} \quad \xrightarrow{s=x} \quad \xrightarrow{s=0}$

$$[y(s)]_{s=0}^{s=x} = y(s) - y(0) = \underbrace{y(x) - 3}_{\text{kezdeti felt. használ}} \rightarrow$$

$$\rightarrow y(x) = 3 + \int_0^x 2y(s) + s - 2 \, ds$$

Picard-föle integrálegyenlet

$$\boxed{2.} \quad y_0(x) = y(0) = 3$$

$$y_1(x) = 3 + \int_0^x 2y_0(s) + s - 2 \, ds$$

$$\vdots$$

$$y_{n+1}(x) = 3 + \int_0^x 2y_n(s) + s - 2 \, ds$$

$$y_1(x) = 3 + \int_0^x 2 \cdot 3 + s - 2 \, ds = 3 + \int_0^x s + 4 \, ds = \left[\frac{s^2}{2} + 4s \right]_0^x + 3$$

$$y_2(x) = 3 + \int_0^x 2 \left(\frac{s^2}{2} + 4s + 3 \right) + s - 2 \, ds = 3 + \int_0^x s^2 + 9s + 4 \, ds =$$

$$= 3 + \left[\frac{s^3}{3} + \frac{9}{2}s^2 + 4s \right]_0^x$$

$$y_3(x) = 3 + \int_0^x 2 \left(\frac{s^3}{3} + \frac{9}{2}s^2 + 4s \right) + s - 2 \, ds = 3 + \int_0^x \frac{2}{3}s^3 + 9s^2 + 9s$$

$$+ 4 \, ds = \left[\frac{2}{12}s^4 + \frac{9}{3}s^3 + 9s^2 + 4s \right]_0^x$$

$\vdots$

$$y_n = \left[\frac{2^{n-2}}{3 \cdot 4 \cdot \dots \cdot (n+1)} s^{n+1} + \dots \right]$$

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$$(1) y' = 2y + x^2$$

$$y(0) = 4$$

$$y = 4 + \int_0^x 2y(s) + s^2 ds \quad \text{Picard}$$

$$y_0(x) = 4 \quad \text{is} \quad y_{n+1}(x) = 4 + \int_0^x 2y_n(s) + s^2 ds$$

$$y_1(x) = 4 + \int_0^x 2 \cdot 4 + s^2 ds = 4 + \left[8s + \frac{s^3}{3} \right]_0^x$$

$$y_2(x) = 4 + \int_0^x 2y_1(s) + s^2 ds = 4 + \int_0^x 2 \left(8s + \frac{s^3}{3} \right) + s^2 ds =$$

$$= 4 + \left[16 \frac{s^2}{2} + 2 \frac{s^4}{4 \cdot 3} + \frac{s^3}{3} + 8s \right]_0^x$$

$$y_3(x) = 4 + \int_0^x 2 \left(16s + 8 \frac{s^2}{3} + \frac{s^3}{3} + \frac{2s^4}{3 \cdot 4} \right) + s^2 ds$$

$$y_n(x) = 4 + \frac{s^3}{3} + \int_0^x 2y_{n-1}(s) ds = 4 + \frac{s^3}{3} + 2 \int_0^x y_{n-1}(s) ds$$

$$= 4 + \frac{s^3}{3} + 2 \int_0^x \left(4 + \int_0^s 2y_{n-1}(t) + t^2 dt \right) ds =$$

$$= 4 + \frac{s^3}{3} + 8s + \int_0^x \int_0^s 4 + 2y_{n-1}(t) + t^2 dt ds = 4 + \frac{s^3}{3} + 8s + 2 \int_0^x \int_0^s \left(\int_0^t 2y_{n-1}(s) + s^2 ds \right) +$$

$$+ \frac{x^4}{2 \cdot 3}$$

$$y(x) = 4 + 8x + 8x^2 + \frac{17}{2^2} \sum_{k=2}^{\infty} \frac{2^k x^k}{k} = 4 + 8x + 8x^2 + \frac{17}{2} (e^{2x} - 1 - 2x - 2x^2)$$

ima mr. $y' = 2y + x^2$

homogén: $y' = 2y \rightarrow \frac{y'}{y} = 2 \rightarrow \log(y) = 2x + C \rightarrow y = C e^{2x}$

$$\underbrace{y' e^{-2x} - 2y e^{-2x}}_{(y e^{-2x})'} = x^2 e^{-2x}$$

$$y e^{-2x} = \int x^2 e^{-2x} = -\frac{1}{2} x^2 e^{-2x} + \int 2x \left(\frac{1}{2} e^{-2x} \right) dx =$$

$$= -\frac{1}{2} x^2 e^{-2x} - \frac{1}{2} x e^{-2x} + \int \frac{1}{2} e^{-2x} dx = -\frac{1}{2} e^{-2x} (x^2 + x) + \frac{1}{2} \left(-\frac{1}{2} \right) e^{-2x} + C$$

$$y = -\frac{x^2}{2} + \frac{x}{2} + \frac{1}{4} + C$$

↑ próbefüggvény

Taylor-sor-módszer

$y' = 2y + x^2$, ha $x = 0$ $y'(0) = 2y(0) + 0 = 8$
 deriváljuk DE-t és behelyettesítjük 0-t

$$y'' = 2y' + 2x \xrightarrow{x=0} y''(0) = 16$$

$$y^{(3)} = 2y'' + 2 \xrightarrow{x=0} y^{(3)}(0) = 34$$

$$y^{(4)} = 2y^{(3)} \xrightarrow{x=0} y^{(4)}(0) = 2 \cdot 34 = 2^2 \cdot 17$$

$$y^{(n)} = 2y^{(n-1)} \rightarrow y^{(n)}(0) = 2y^{(n-1)}(0) = \dots = 2^{n-3} y'''(0) = 2^{n-2} \cdot 17$$

Taylor-sor képlete

$$y(x) = \sum_{k=0}^{\infty} \frac{y^{(k)}(0)}{k!} (x-x_0)^k = y(0) + y'(0)x + \frac{y''(0)}{2} x^2 + \sum_{k=3}^{\infty} \frac{y^{(k)}(0)}{k!} x^k = 4 + 8x + 8x^2 + \sum_{k=3}^{\infty} \frac{2^{k-2} \cdot 17}{k!} x^k = \dots =$$

$$= -\frac{1}{2}x^2 - \frac{1}{2}x - \frac{1}{4} + \frac{17}{4}e^{2x}$$

Határozatlan együtthatós módszer

$$y = C_0 + C_1(x-x_0) + C_2(x-x_0)^2 + \dots + C_k(x-x_0)^k + \dots \quad \text{alakin megoldást keresünk}$$

$$\rightarrow y' = C_1 + 2C_2(x-x_0) + \dots + kC_k(x-x_0)^{k-1} + \dots$$

// konvergenciasugaron belül lehet deriválni

$$y(x_0) = C_0, \text{ ha } x_0 = 0, \text{ akkor } C_0 = y(0) = 4 \text{ és a DE}$$

$$C_1 + 2C_2x + \dots + kC_kx^{k-1} + \dots = (C_0 + 2C_1x + \dots + 2C_kx^2 + \dots) + x^2$$

$$C_1 = 2C_0$$

$$2C_2 = 2C_1 = 16$$

$$3C_3 = 2C_2 + 1 = 17$$

$$4C_4 = 2C_3 \rightarrow C_4 = \frac{2 \cdot 17}{3 \cdot 4}$$

$$\vdots$$

$$nC_n = 2C_{n-1} \rightarrow C_n = 2^{n-2} \cdot \frac{17}{n!}$$

$$y(x) = 4 + 8x + 8x^2 + \sum_{k=3}^{\infty} \frac{2^{k-2} \cdot 17}{k!} x^k$$

~~same story as before~~
oh shit, here we go again

② $y' = y^4 \cos(x) + y \operatorname{tg}(x) \quad / y^4 \quad (\text{Bernoulli-féle DE})$

$$\frac{y'}{y^4} = \cos(x) + \underbrace{y^{-3}}_u \operatorname{tg}(x) \quad \left\| \begin{array}{l} u(x) = [y(x)]^{-3} \\ u' = -3 y^{-4} \cdot y' \end{array} \right.$$

$$-3 y^{-4} y' = \cos(x) + y^{-3} \operatorname{tg}(x)$$

$$u' = -3 \cos(x) - 3 u \operatorname{tg}(x)$$

homogén

$$\frac{u'}{u} = -3 \operatorname{tg}(x) \rightarrow \log(x) = -3 \int \frac{\sin x}{\cos x} = 3 \log(\cos x) + c$$

$$x = e^{\cos^3(x)} \cdot C_2$$

$$\underbrace{u' \cos^3(x) + 3u \operatorname{tg}(x) \cdot \cos^3(x)}_{u \cos^3 x} = -3 \cos^2(x)$$

$$u \cos^3 = \int -3 \cos^2(x) = -3 \operatorname{tg}(x) + c$$

$$u = -3 \sin(x) \cos^2(x) + c \cos^3(x)$$

$$y = u^{-\frac{1}{3}} = \left(-3 \sin(x) \cos^2(x) + c \cos^3(x) \right)^{-\frac{1}{3}} \quad \text{és } \overset{\text{függvény}}{y=0}$$

1. $y' = \frac{y^2}{\cos(x)} - y \operatorname{tg}(x) - \cos(x)$ emiatt nem Bernoulli

$y' = a(x)y^2 + b(x)y + c(x)$ Riccati-egyenlet

y_0 megoldás, hh

$$y_0' = a(x)y_0^2 + b(x)y_0 + c(x)$$

$$(y - y_0)' = a(x)(y^2 - y_0^2) + b(x)(y - y_0)$$

$$\underbrace{(y - y_0)}_z (y + y_0) =$$

$y' = a(x)z(z + 2y_0) + b(x)z$ Bernoulli-féle DE $z = y - y_0$

$y = y_0 + z$

$$\cos(x) = \frac{\cos^2}{\cos} - \cos \operatorname{tg} - \cos = -\sin$$

$$y = z + \cos(x)$$

$$z' - \sin(x) = \frac{(z + \cos(x))^2}{\cos(x)} - (z + \cos(x)) \operatorname{tg}(x) - \cos(x)$$

$$z' - \sin(x) = \frac{z^2 + 2\cos(x)z + \cos^2(x)}{\cos(x)} - z \operatorname{tg}(x) - \sin(x) - \cos(x)$$

$$\frac{z'}{z^2} = \frac{1}{\cos(x)} + \frac{1}{z}(2 - \operatorname{tg}(x)) \quad / (-1)$$

$$-\frac{z'}{z^2} = -\frac{1}{\cos(x)} + \frac{1}{z}(\operatorname{tg}(x) - 2)$$

$$u = \frac{1}{z} \quad u' = \frac{z'}{z^2}$$

$$u' = \frac{-1}{\cos^2(x)} + u(\operatorname{tg}(x) - 2)$$

homogén: $u'(\operatorname{tg}(x) - 2)$

$$\frac{u'}{u} = \operatorname{tg}(x) - 2$$

$$\begin{aligned} \log(u) &= -\log(\cos(x)) - 2x + C \rightarrow u_h = C e^{-2x} \cdot \cos^{-1}(x) = \\ &= C \frac{e^{-2x}}{\cos(x)} \end{aligned}$$

$$u' + u(2 - \operatorname{tg}(x)) = -\frac{1}{\cos(x)} \quad / \cdot e^{2x} \cos(x)$$

$$e^{2x} \cos(x) (u' + u(2 - \operatorname{tg}(x))) = -e^{2x}$$

$$e^{2x} \cos(x) (u' + u(2 - \operatorname{tg}(x))) = -e^{2x}$$

$$(e^{2x} \cos(x) u)' = -e^{2x} \quad / \int dx$$

$$e^{2x} \cos(x) \cdot u = -\frac{1}{2} e^{2x} + C$$

$$u = -\frac{1}{2 \cos(x)} + \frac{C}{e^{2x} \cos(x)} = \frac{-e^{2x} + 2C}{2e^{2x} \cos(x)}$$

$$y = \frac{1}{u} = \frac{2e^{2x} \cos(x)}{-e^{2x} + 2C} \rightarrow y = \frac{2}{-1 + 2C e^{-2x}} + \cos(x) =$$

$$= \frac{2e^{2x} \cos(x)}{-e^{2x} + 2C} + \cos(x)$$

lineáris
Bernoulli SZEP
Piccarti

→ menjünk tovább

Azt az
ország köny-
vet, amit
ajánlott
el tudod
mutatni?

$$(2) a) y' = y^2 \rightarrow -\frac{y'}{y^2} = -1 \xrightarrow{\int} \frac{1}{y} = -x + c \rightarrow y = \frac{1}{c-x}$$

$$b) y' = \frac{1}{y^2} \rightarrow 3y^2 y' = 3 \xrightarrow{\int} y^3 = 3x + c \rightarrow y = \sqrt[3]{3x+c}$$

$$(3) xy' = y^2 + 4y \text{ szétválasztható}$$

$$\frac{y'}{y^2+4y} = \frac{1}{x}$$

|| nézzük meg a konstans megoldásokat

$$\frac{\frac{dy}{dx}}{y^2+4y} = \frac{1}{x} \rightsquigarrow \int \frac{dy}{y^2+4y} = \int \frac{dx}{x} \rightsquigarrow \int \frac{1}{y(y+4)} dy =$$

$$= \frac{1}{4} \int \frac{1}{y} - \frac{1}{y+4} dy = \frac{1}{4} (\log(y) - \log(y+4)) + c = \frac{1}{4} \log\left(\frac{y}{y+4}\right) + c$$

$$\log\left(\frac{y}{y+4}\right) = 4 \log(x) + 4c$$

$$\frac{y}{y+4} = Cx^4 \rightarrow y = \frac{4Cx^4}{1-Cx^4} \text{ és } y = -4$$

$$(4) \text{ Egyöntetű fokszámú DE}$$

$$y' = f\left(\frac{y}{x}\right) \rightarrow u(x) = \frac{y(x)}{x} \text{ -et kell helyettesíteni}$$

$$y' = (ux)' = y' \cdot x + u = f(u)$$

$$u' \cdot x = f(u) - u \text{ szétválasztható}$$

$$y' = \frac{2x+y}{x-y} \quad y = ux$$

$$u'x + u = \frac{2x+ux}{x-ux} = \frac{2+u}{1-u} \rightarrow u'x = \frac{2+u}{1-u} - u = \frac{2+u^2}{1-u}$$

11

$$\frac{1-u}{2+u^2} u' = \frac{1}{x}$$

$$\int \frac{1-u}{2+u^2} du = \int \frac{dx}{x}$$

$$\int \frac{1}{2+u^2} du = \int \frac{\frac{1}{2}}{\frac{1}{2}u^2+1} du = \frac{\sqrt{2}}{2} \operatorname{arctg}\left(\frac{u}{\sqrt{2}}\right) + C$$

$$\int \frac{u}{2+u^2} du = \frac{1}{2} \int \frac{2u}{2+u^2} du = \frac{1}{2} \log(u^2+2) + C$$

$$\frac{\sqrt{2}}{2} \operatorname{arctg}\left(\frac{u}{\sqrt{2}}\right) - \frac{1}{2} \log(u^2+2) = \log(x) + C$$

$$e^{\frac{\sqrt{2}}{2} \operatorname{arctg}\left(\frac{u}{\sqrt{2}}\right)} e^{-\frac{1}{2} \log(u^2+2)} = \frac{e^{\frac{\sqrt{2}}{2} \operatorname{arctg}\left(\frac{u}{\sqrt{2}}\right)}}{\sqrt{e^{u^2+2}}} = Cx \rightarrow$$

delete
stroke

$$\rightarrow e^{\frac{\sqrt{2}}{2} \operatorname{arctg}\left(\frac{u}{\sqrt{2}}\right)} = C(y^2+2x^2)$$

5. $y' = f(ax+b)$

$$u' = a + by' \rightarrow y' = \frac{u'-a}{b}$$

$$\frac{u'-a}{b} = f(u) \text{ szétválasztható}$$

$$(1-2x-2y)y' = x+y+1$$

$$a=1, b=1, f=x+1$$

$$y' = \frac{x+y+1}{1-2x-2y}$$

$$u = x+y$$

$$(1-2u)(u'-1) = u+1$$

$$(1-2u)u' = 2-u$$

$$(u=2 \text{ mo})$$

$$\frac{(1-2u)u'}{2-u} = 1$$

$$\rightarrow \int \frac{1-2u}{2-u} du = \int dx$$

$$\int \frac{1-2u}{2-u} = \int \frac{4-2u}{2-u} - \frac{3}{2-u} = \int 2 + \int \frac{3}{2+u} = 2u + 3 \log(u-2) = x + C$$

$$e^{2u} \cdot (u-2)^3 = c e^x$$

$$e^{2(x+y)} (x+y-2)^3 = c e^x$$

$$(6.) \quad x^3(y' - x) = y^2$$

$$x^3 y' - x^4 = y^2$$

$y = z^m$ -nel lehet-e homogén feltevéssel
tenni?

$$y' = m z^{m-1}$$

$$m=2$$

$$x^3 \left(\left(\frac{y}{x} \right)' - 1 \right) = \frac{y^2}{x^4}$$

$$x^3 \frac{y'}{x} - x^4 = \frac{y^2}{x^4}$$

$$x^3 \frac{y'}{x} = \frac{y^2}{x^4} + x^4$$

$$z' = \frac{z^4 + x^4}{x^3 z} = \frac{\left(\frac{z}{x}\right)^4 + 1}{2\left(\frac{z}{x}\right)}$$

$$z = ux$$

$$z' = u'x + u$$

$$u'x + u = \frac{u^4 + 1}{2u}$$

$$u'x = \frac{u^4 + 1}{2u} - u = \frac{u^4 + 1 - 2u^2}{2u} = \frac{(u^2 - 1)^2}{2u}$$

$$\int \frac{2u}{(u^2 - 1)^2} du = \int \frac{1}{x} dx$$

$$\frac{f'}{f^2} - \frac{1}{(u^2 - 1)} = \log(x) + C \rightarrow e^{\frac{x^2}{x^2 - y}} = Cx$$

+ ellenőrizni a szabálytalan lépésekből szárm.
megoldást

11# a) $y' + \frac{y}{x} = x^2 y^4$ Bernoulli

b) $y' = \frac{y^2}{x^2} - 2 \frac{y}{x}$ Egyöntetű együtthatójú fokozatú

c) $4x - 3y + y'(2y - 3x) = 0$ Egyöntetű fokozatú

d) $e^{4x+y} - \frac{1}{2} = y'$

e) $e^{2x+\frac{y}{2}} - \frac{1}{2} = y'$

f) $y' = \frac{5xy + 2y^2}{x^2}$ (lehet, hogy nem jó)

Ricatti / homogén egyöntetű fokozatú

g) légitellenállás (szétválasztható változójú)

lineáris

↓
Bernoulli

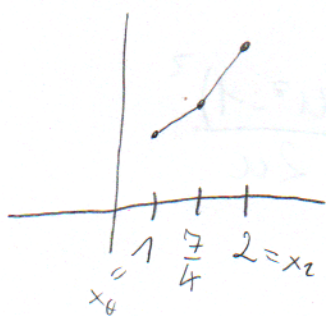
↓
Ricatti → inkább máshogy, mert gusztustalan

↓
szétválasztható változójú

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1. $y' = x^2 - y^2 + 1$ $x_0 = 1$ $x_1 = \frac{7}{4}$ $x_2 = 2$
 $y(1) = 1$

Számítsuk ki $y(2)$ közelítését az EE és az IE módszerrel is



EE $y_{k+1} = y_k + (x_{k+1} - x_k) \underbrace{f(x_k, y_k)}_{y'(x_k, y_k)}$

IE $y_0 = 1$

$y_1 = y_0 + (x_1 - x_0) f(x_0, y_0) =$

$= 1 + \frac{3}{4} (1^2 - 1^2 + 1) = \frac{7}{4}$

$y_2 = y_1 + (x_2 - x_1) f(x_1, y_1) =$

$= \frac{7}{4} + \frac{1}{4} \left(\left(\frac{7}{4} \right)^2 - \left(\frac{7}{4} \right)^2 + 1 \right) = 2$

nem meglepő (mert $y=x$ a megoldás)

$$IE \quad y_{k+1} = y_k + (x_{k+1} - x_k) f(x_{k+1}, y_{k+1})$$

$$y_0 = 1$$

$$y_1 = y_0 + (x_1 - x_0) f(x_1, y_1) = 1 + \frac{3}{4} \left(\left(\frac{7}{4} \right)^2 - y_1^2 + 1 \right)$$

$$64y_1 = 64 + 3(49 - 16y_1^2 + 16)$$

$$48y_1^2 + 64y_1 - 259 = 0 \quad y_1 \in \left\{ \frac{-64 \pm \sqrt{64^2 + 4 \cdot 48 \cdot 259}}{96} \right\}$$

$$\frac{-8 \pm \sqrt{64 + 3 \cdot 259}}{12} = \frac{-8 \pm 29}{12} \rightarrow \frac{21}{12} = \frac{7}{4}$$

valaszul
ezt

$$y_2 = y_1 + (x_2 - x_1)(x_2^2 - y_2^2 + 1)$$

$$y_2 = 7 + 5 - y_2^2 =$$

$$y_2^2 + 4y_2 + 12 = (y_2 + 6)(y_2 - 2) \rightarrow \begin{matrix} 2 \\ -6 \end{matrix}$$

korábban tudjuk,
hogy

|| ha az ~~valamely~~ meren rezgéseknél az IE
az IE biztonságosabb

2. $e^{-y} dx - (2y + x e^{-y}) dy = 0$

$$H(x, y) \quad dH = \frac{\partial H}{\partial x} dx + \frac{\partial H}{\partial y} dy$$

$$M dx + N dy = 0 \quad \text{egzakts, ha } \exists H: M = \frac{\partial H}{\partial x} \quad N = \frac{\partial H}{\partial y}$$

egzakts DE
 $dH = 0$
 $H = \text{állandó}$

ha van ilyen, hkk

$$\frac{\partial M}{\partial y} = \frac{\partial^2 H}{\partial y \partial x} = \frac{\partial^2 H}{\partial x \partial y} = \frac{\partial N}{\partial x}$$

Egzaktság ellenőrzése:

$$\frac{\partial M}{\partial y} = -e^{-y} = \frac{\partial N}{\partial x} \quad \# \text{ egzakt}$$

$$H = \int M dx = \int e^{-y} dx = x e^{-y} + c(y) \quad / \frac{\partial}{\partial y}$$

$$H'_y \stackrel{!}{=} N = -x e^{-y} + c'(y)$$

$$\begin{aligned} & -2y - x e^{-y} \\ & -2y = c'(y) \quad / \int dy \end{aligned}$$

$$-y^2 + c = c(y) \rightarrow H(x, y) = x e^{-y} - y^2 + c$$

a megoldás: $x e^{-y} - y^2 = \text{állandó}$

$$(3.) \quad a) \quad \overbrace{2x}^M dx + \overbrace{2y}^N dy = 0$$

$$\underline{H(x, y) = x^2 + y^2 = \text{állandó}}$$

$$\frac{\partial M}{\partial y} = 0 = \frac{\partial N}{\partial x}$$

$$b) \quad \overbrace{y \cos(xy)}^M dx + \overbrace{x \cos(xy)}^N dy = 0$$

$$\frac{\partial M}{\partial y} = -x y \sin(xy) = \frac{\partial N}{\partial x}$$

$$\sin(xy) = c$$

$$c) \quad \frac{y}{x} dx + (\log(x) + 2y) dy = 0$$

$$y \log(x) + y^2 = c$$

$$(4) \quad \underbrace{(x^2 + y^2 + x)}_M dx + \underbrace{y^2}_N dy$$

$$M \quad \frac{\partial M}{\partial y} = 2y \neq \frac{\partial N}{\partial x}$$

keresünk egy $m(x, y)$ multiplikátort (integráló tényező),
amivel beszorozva egzakttá válik

$$M dx + N dy = 0 \quad / \cdot m$$

$$mM dx + mN dy = 0$$

ha egzakt, akkor

$$(mM)'_y = (mN)'_x$$

az egyszerű DE-re m -re, általában sokkal
nehézebb megoldani. Ezért feltesszük, hogy $m(x, y) = \varphi(z)$
csak z -től, az (x, y) valamely egyszerű kifejezésétől függ

$$\text{Lehetőségek: } z \in \{x, y, x+y, xy, \frac{x}{y}\}$$

$$m'_y M + m M'_y = m'_x N + m N'_x$$

$$\varphi' \cdot \frac{\partial z}{\partial y} \cdot M + \underbrace{m}_{\varphi} M'_y = \varphi' \cdot \frac{\partial z}{\partial x} N + \underbrace{m}_{\varphi} N'_x$$

$$\varphi' \left(M \frac{\partial z}{\partial y} - N \frac{\partial z}{\partial x} \right) = \varphi (-M'_y + N'_x)$$

Alapon $\triangleleft$

$$\boxed{\frac{\varphi'}{\varphi} = \frac{-M'_y + N'_x}{M \frac{\partial z}{\partial y} - N \frac{\partial z}{\partial x}}}$$

ha a jobb oldal csak
 z -től függ, akkor z meghat.
és van z -től függő multipli-
kátor

miszt a feladathoz:
Van-e x-től függő multiplikátor?

$$\frac{\varphi'}{\varphi} = \frac{2y+0}{M \cdot 0 - N \cdot 1} = \frac{-2y}{-y} = +2 \quad \text{csak } x\text{-től függ}$$

$$\log(\varphi) = \int -2dx = -2x (+c_1)$$

$$\varphi = e^{+2x} (c_2)$$

|| $z = xy$ eset

$$\frac{\varphi'}{\varphi} = \frac{-2y}{Mx - Ny} = \frac{-2y}{(x^2 + y^2 + x)x - y^2} \quad \text{nem csak } z = xy \text{ függvénye}$$

$$e^{+2x}(x^2 + y^2 + x)dx + e^{+2x}ydy = 0$$

ellenőrizzük,
hogy tényleg egyenlet-e
!!!
ooo

$$H = \int e^{2x} y dy = \frac{1}{2} e^{2x} y^2 + c(x) \quad \left/ \frac{\partial}{\partial x} \right.$$

$$e^{2x}(x^2 + y^2 + x) = e^{2x} y^2 + c'(x)$$

$$\int e^{2x}(x^2 + x) = \int c'(x) dx$$

$$c_2 + \frac{1}{2} x^2 e^{2x} = c(x)$$

$$\frac{1}{2} (x^2 + y^2) e^{2x} = \text{áll}$$

$$(x^2 + y^2) e^{2x} = \text{áll}$$

$$(5.) \quad (1 + y^2 \sin(2x)) dx + (2y \cos^2(x)) dy = 0$$

$$\frac{\partial M}{\partial y} = 2y \sin(2x) = \frac{\partial N}{\partial x} = 2y \cdot 2 \cos(x) \sin(x)$$

tehát egyenlő

$$(6.) \quad y^2 dx + (xy^4 + xy \log(x)) dy = 0$$

$$\frac{\partial M}{\partial y} = 2y$$

$$\frac{\partial N}{\partial x} = x$$

$$\frac{\partial N}{\partial x} = y^4 + y \log(x) + y$$

$$\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} = y^4 + y \log(x) - y$$